Correlation and Regression

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What is correlation and correlation co-efficient?

The relationship between two or more variable is called correlation. The primary objective of correlation analysis is to measure the strength or degree of relationship between two or more variables. It the change in one variable effects a change in the other variable, the variable are said to be correlated.

Method of studing simple correlation:-

1. Scatter Diagram method.
2. Karl Pearson's coefficient of correlation.
3. Spearman's Rank correlation.

Correlation coefficient:-

The mathematical measures of correlation are called correlation coefficient (r_{xy}/r).

$$\boxed{r_{xy} = \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}}}$$

This formula was given by Karl pearson's in 1890.

Assumptions of Pearson's correlation coefficient:-

1. There is a linear relationship between two variables when the two variables are plotted on a scatter diagram. a straight line will be formed by the points.
2. Cause and effect relation exists between different forces operating on the item of the two variable series.

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consider the following data

x	y
1	6
2	4
3	3
4	5
5	4
6	2

Here,

$\sum x = 21, \sum y = 24, \sum x^2 = 91, \sum y^2 = 106, \sum xy = 75,$

$n = 6$

We know,

$$r_{xy} = \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}}$$

$$= \frac{6\times 75 - 21\times 24}{\sqrt{\{6\times 91 - (21)^2\}\{6\times 106 - (24)^2\}}}$$

$$= \frac{-54}{\sqrt{105\times 60}}$$

$= -0.680$ (Ans)

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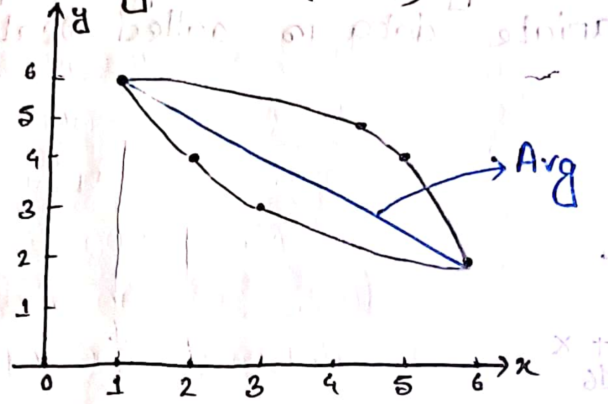
Scatter Diagram :- (বিক্ষিপ্ত)



#consider the following datas :-

x	y
11	16
12	14
13	13
14	13
15	14
16	12

Here,

$\sum x = 81,\ \sum y = 84,\ \sum x^2 = 1111,\ \sum y^2 = 1186,\ \sum xy = 1125.$

$n = 6$

we know,

$$r_{xy} = \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}}$$

$$= \frac{6\times1125 - 81\times84}{\sqrt{\{6\times1111 - (81)^2\}\{6\times1180 - (84)^2\}}}$$

$$= \frac{-54}{\sqrt{105\times2460}} = -0.68$$ (Ans).

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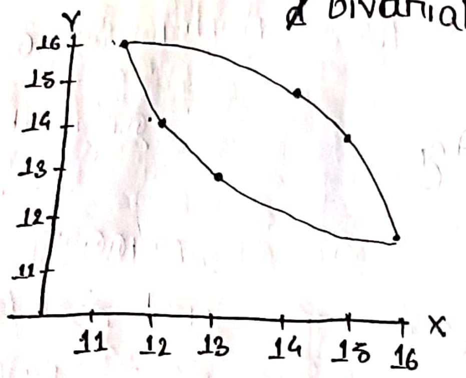
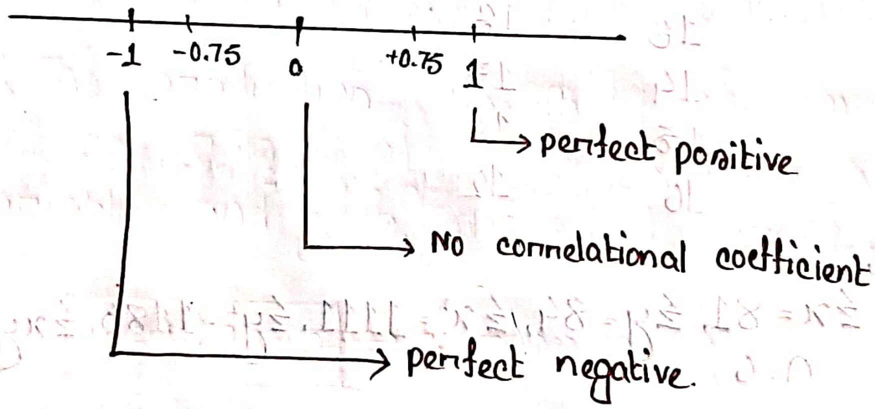
Scatter Diagram:- The diagrammetic way of representing ~~a~~ bivariate data is called scatter diagram.



$$\boxed{-1 \leq r < +1}$$



① $r=0 \longrightarrow$ x and y are independent. There is no correlation coefficient.

② $0 < r_{xy} < 0.75 \longrightarrow$ Positive correlation coefficient.

③ $0.75 \leq r < 1 \longrightarrow$ Strongly positive correlation coefficient.

④ $r=1 \longrightarrow$ Perfect positive correlation coefficient.

Intercepts of r:-

$r = +1 \longrightarrow$ a perfect positive relationship between x and y.

$r = -1 \longrightarrow$ a perfect negative " between x and y.

$r = 0 \longrightarrow$ there is no linear relationship between x and y. Here two variables are linearly independent.

$0 < r < 1 \longrightarrow$ a positive relationship between x and y.

$-1 < r < 0 \longrightarrow$ a negative relationship between x and y.

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⑤ $-0.75 < r < 0$ ⟶ negative correlation coefficient.

⑥ $-1 < r \leq -0.75$ ⟶ Strongly negative correlation coefficient

⑦ $r = -1$ ⟶ Perfectly negative correlation coefficient

Comment for the previous problem:-

Ans. $r_{xy} = 0.68$.

The correlation between x and y is negative.

Properties of correlation coefficient:-

1. Correlation coefficient is independent of change of origin and scale measurement.

2. Correlation coefficient lies between -1 to +1.

i.e $\boxed{-1 < r_{xy} < +1}$

3. Correlation coefficient is symmetric,

i.e. $\boxed{r_{xy} = r_{yx}}$

4. Correlation coefficient is the geometric mean of regression coefficient

i.e $\boxed{r_{xy} = \sqrt{b_{xy} \cdot b_{yx}}}$

5. For two independent variables correlation coeffic

is zero.

6. Correlation coefficient is unit free.

What is Regression and what is regression coefficient ?

Regression is the ~~fundame~~ functional relationship between two variables and of the two variables one may represent cause and other may represent effect.

This term was used by a famous Biometrician Sir F. Galton in 1877.

Example:- The production of paddy of amount y is dependent on rainfall of amount x. Here x is independent variable and y is dependent variable.

$$\boxed{y = a + bx + e} \rightarrow \text{regression eq}^n$$

Here,

y = Dependent variable / Predicted variable

a = constant / intercept term.

b = regression coefficient.

x = Independent variable / Predictor variable / regressor

e = error term.

The yield of paddy depends on water, fertilizers and irrigation etc.

$$y = a + \underbrace{b_1 x_1}_{\text{water}} + \underbrace{b_2 x_2}_{\text{fertilizers}} + \underbrace{b_3 x_3}_{\text{Irrigation}} + e$$

→ The mathematical measures of regression are called the coefficient of regression.

$$\boxed{b_{yx} = \frac{n\Sigma xy - \Sigma x . \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}} \rightarrow \begin{array}{l} y = \text{dependent} \\ x = \text{Independent.} \end{array}$$

$$\boxed{b_{xy} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma y^2 - (\Sigma y)^2}} \rightarrow \begin{array}{l} x = \text{dependent} \\ y = \text{Independent} \end{array}$$

Let (x_1, y_1) (x_2, y_2) - - - (x_n, y_n) be the pairs of n observations, Then the regression co-efficient y on x denoted by b_{yx} and regression co-efficient x on y.

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Show that $r_{xy} = \sqrt{b_{xy} \cdot b_{yx}}$:

We know,

$$b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \quad \&$$

$$b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}.$$

$\therefore$ R.H.S $= \sqrt{b_{yx}\, b_{xy}}$

$$= \sqrt{\frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \cdot \frac{n\sum xy - \sum x . \sum y}{n\sum y^2 - (\sum y)^2}}$$

$$= \sqrt{\frac{(n\sum xy - \sum x \sum y)^2}{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}}$$

$$= \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}}}$$

$= r_{xy}$.

$=$ L.H.S.

$\therefore r_{xy} = \sqrt{b_{yx} . b_{xy}}$. (Shown)

Properties of Regression:-

1. Regression coefficient is independent of change of origin but not of scale.

2. Regression coefficient lies between $-\infty$ to $+\infty$

i.e $\boxed{-\infty < b_{xy}/b_{yx} < +\infty}$

3. Regression isn't symmetric.

i.e $\boxed{b_{xy} \neq b_{yx}}$

4. ~~Reasi~~ The geometric mean of regression is equal to the correlation coefficient.

i.e, $\boxed{r_{xy} = \sqrt{b_{yx} \cdot b_{xy}}}$

5. The arithmetic mean of regression coefficient is greater than correlation coefficient.

i.e $\boxed{r_{xy} \leq \frac{b_{xy} + b_{yx}}{2}}$

6. Regression coefficient is not unit free.

◫ Difference between the regression coefficient & correlation coefficient:-

Correlation coefficient	Regression coefficient
1. It lies between -1 to +1.	1. It lies between $-\infty$ to $+\infty$.
2. It ~~ay~~ is symmetric. $r_{xy} = r_{yx}$	2. It is not symmetric $b_{xy} \neq b_{yx}$
3. For two independent variable correlation coefficient is zero.	3. –

Correlation coefficient	Regression coefficient
1. The numerical value by which we measure the strength of linear relationship between two or more variable is called correlation coefficient.	1. The mathematical measures of regression are called the coefficient of regression.
2. Correlation coefficient is independent of change of origin and scale of measurement.	2. Regression coefficient is independent of change of origin but not scale.
3. It lies between -1 to +1 $-1 \leq r_{xy} \leq +1$	3. It lies between $-\infty$ to $+\infty$ $-\infty < b_{xy} / b_{yx} < +\infty$

4. It is symmetric. i.e $r_{xy} = r_{yx}$.	4. It is not symmetric. i.e $b_{yx} \neq b_{xy}$.
5. It is always unit free.	5. It is not pure number.
6. When $r=0$ then the variables are connected.	6. When $r=0$, then the two lines of regression are perpendicular to each other.

☐ Regression Equation:-

The regression Equation of y on x is expressed as follows -

y = Dependent variable
x = independent variable
a = intercept term
b = slope of the line.

$$y = a + bx$$
$$a = y - bx$$
$$= \bar{y} - b\bar{x}$$

and $b_{yx} = \dfrac{n\sum xy - \sum x \sum y}{\{n\sum x^2 - (\sum x)^2\}}$

$$a = \bar{y} - b\bar{x}$$
$$= \frac{\sum y}{n} - b\frac{\sum x}{n}$$

Similarly, the regression equation of x on y is expressed as follows:

y = Independent variable
x = Dependent variable
a = intercept term
b = slope of the line

$$x = a + by$$
$$\therefore a = x - by$$
$$= \bar{x} - b\bar{y}$$

and $b_{xy} = \dfrac{n\sum xy - \sum x \sum y}{\{n\sum y^2 - (\sum y)^2\}}$

$$a = \bar{x} - b\bar{y}$$
$$= \frac{\sum x}{n} - b\frac{\sum y}{n}.$$

Prob'- 1

Temp of water	68	65	70	62	60	55	58	65	69	63
Reduction pulse rate	2	5	1	10	9	13	10	3	4	6

Sol:-

x	y	x^2	y^2	xy
68	2	4624	4	136
65	5	4225	25	325
70	1	4900	1	70
62	10	3844	100	620
60	9	3600	81	540
55	13	3025	169	715
58	10	3364	100	580
65	3	4225	9	195
69	4	4761	16	276
63	6	3969	36	378

Here, $\Sigma x = 635$, $\Sigma y = \overset{63}{\cancel{72}}$, $\Sigma x^2 = 40537$, $\Sigma y^2 = 541$, $\Sigma xy = 3835$

$\therefore$ Correlation coefficient,

$$r_{xy} = \frac{n\Sigma xy - \Sigma x . \Sigma y}{\sqrt{\{n\Sigma x^2 - (\Sigma x)^2\}\{n\Sigma y^2 - (\Sigma y)^2\}}}$$

$[n = 10]$

$$= \frac{10.3835 - 635 \times \cancel{72}\,63}{\sqrt{\{10.40537 - (635)^2\}\{10.541 - (63)^2\}}}$$

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$= -0.94$. (Ans)

The result -0.94 indicated that the correlation coefficient between temperature of water and reduction in pulse rate is highly negatively corrected.

Prob-2

x	1	2	3	4	5	6
y	6	4	3	5	4	2

i. Find r_{xy}.

ii. Draw Scatter Diagram.

iii. Estimate the value of y when $x=3$.

iv. Calculate regression line of x on y.

v. Predict the value of x when $y=4.5$

Sol:-

iii. $\Sigma x = 21$, $\Sigma y = 24$, $\Sigma x^2 = 91$, $\Sigma y^2 = 106$. $\Sigma xy = 75$

the regression eq^n of y on x is

$$y = a + bx \quad \text{--- (i)}$$

Here,

$$a = \bar{y} - b\bar{x}$$

$$= \frac{\Sigma y}{n} - b\frac{\Sigma x}{n}$$

$$= \frac{24}{6} - b\frac{21}{6} \quad [n=6] \quad \text{--- (ii)}$$

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and,
$$b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$= \frac{6 \times 75 - 21 \times 24}{6 \times 91 - (21)^2}$$

$$= -0.514.$$

Putting the value of b on eqn (ii),

$$a = \frac{24}{6} - (-0.514)\frac{21}{6}$$

$$= 5.8.$$

Now, putting the value of a and b in eqn (i),

$$y = 5.8 + (-0.514)x$$

$$y = 5.8 - 0.514 \times 3 \quad [\because x = 3)$$

$$= 4.258. \text{ (Ans)},$$

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